

PHD ANNUAL REPORT 1 FOR ICRAR / UWA

FIRST YEAR REVIEW

**A resolved view of star formation, gas cycling and chemical enrichment in
Virgo Cluster spirals**

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Abstract

Galaxy evolution is regulated by the baryon cycle: gas accretion fuels star formation, feedback drives outflows, and recycling redistributes gas and metals. Gas-phase metallicity (Z_{gas}) is a powerful tracer of this regulation because it encodes the competition between enrichment by young stars and dilution/removal by gas flows. A key open question is whether the widely used mass–SFR– Z_{gas} relations primarily emerge from *local* self-regulation within galaxies or instead reflect global and environmental control.

My PhD aims to build a spatially resolved, multi-tracer framework for how gas, stars, and metals co-regulate galaxy evolution in Virgo spirals. Project 1 is complete: my first-author paper **MAUVE–MUSE: When Metallicity Follows or Fights Star Formation—A Mass-Dependent Inversion in Virgo Galaxies** was submitted to MNRAS on 20 Feb 2026. This work shows that the resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling is not universal: its sign and strength can invert in a mass-dependent way, consistent with a transition between enrichment-dominated and gas-supply/dilution-dominated regimes.

I am now advancing to Project 2, which expands the MAUVE–MUSE analysis to the full 3rd internal release (26 galaxies) with homogeneous, value-added *nGIST* products and improved ionized-gas diagnostics. A central goal is to quantify whether, and by how much, environment modulates metallicity and the resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling once measurement systematics (e.g. DIG contamination, ionization conditions, electron density, and T_e -based metallicity where feasible) are controlled. In Project 3, I will add MAUVE–ALMA molecular gas maps to directly test the gas-supply interpretation implied by regulator models, linking Z_{gas} and star formation to the cold-gas fuel reservoir (Σ_{H_2}) and related cycling proxies. Together, these projects will establish a comprehensive, spatially resolved picture of how gas, stars, and metals co-evolve in different physical and environmental regimes.

1 PhD Project Plan

Overall goal: build a spatially resolved, multi-tracer picture of baryon-cycle regulation in Virgo spirals by connecting metallicity, star formation, ionized-gas conditions, and cold-gas supply.

1.1 Project 1 (completed): Resolved metallicity–star-formation coupling in 14 Virgo spirals

Status: Completed; MNRAS submission on 20 Feb 2026.

Although globally-integrated studies often find that, at fixed stellar mass, high star formation rate (SFR) galaxies are relatively metal-poor while lower-SFR systems are more metal-rich, the corresponding coupling between gas-phase metallicity (Z_{gas}) and star formation on sub-galactic scales remains poorly constrained. In this study, we analyse 14 Virgo spiral galaxies from the MAUVE–MUSE survey to revisit the resolved mass–metallicity relation (rMZR) and its secondary dependence on SFR surface density (Σ_{SFR}) at ~ 100 pc scales. We construct co-spatial maps of stellar mass surface density (Σ_*), Σ_{SFR} and gas-phase oxygen abundance and verify our results across several strong-line metallicity diagnostics. MAUVE–MUSE galaxies follow a standard rMZR, but once binned in Σ_* , we find a clear mass-dependent inversion in the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation: low- Σ_* bins show an anti-correlation between Z_{gas} and Σ_{SFR} , whereas high- Σ_* regions show a positive correlation, with an inversion point at $\log_{10}(\Sigma_*/M_{\odot} \text{ kpc}^{-2}) \simeq 7.5$ –

8.0. We find that both correlated and anti-correlated H II regions coexist within the same discs, and that the observed mass dependence emerges only when grouping H II spaxels by Σ_* . To interpret this behaviour, we develop a spatially resolved gas–regulator model and show that the observed correlation and anti-correlation between Z_{gas} and Σ_{SFR} arise from the competition between star-formation-driven and gas-supply-driven variability: the sign of the local $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation is set by which of these dominates. This framework can be naturally extrapolated to the integrated scenario, providing a unified explanation for both the resolved and global $Z_{\text{gas}}-\text{SFR}$ relations.

1.2 Project 2 (current): 26 MAUVE–MUSE galaxies — environment and ionized-gas physics of resolved regulation

Motivation: Project 1 identified a clear, mass-dependent inversion in the resolved $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling that is not obviously driven by environment in the 14-galaxy sample. The next step is to (i) increase statistics and (ii) determine whether, and how, *environment and ionized-gas conditions* modulate metallicity and the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation.

Objectives:

1. **Increase statistics and homogenise measurements:** expand to 26 MAUVE–MUSE galaxies (DR3) with uniform emission-line products and quality control.
2. **Validate Z_{gas} with direct abundances:** obtain T_e -based metallicities where feasible to calibrate/validate strong-line O/H and assess biases across parameter space.
3. **Quantify modulation by environment and ISM conditions:** test whether environment changes (a) the metallicity distribution and (b) the sign/strength of the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ coupling at fixed Σ_* , and assess how this depends on DIG fraction, ionization conditions, and electron density.

Core tasks (deliverables-oriented):

- **Refine *nGIST* pipeline for value-added products:** robust spatial masking (foreground/background), skyline masking, continuum subtraction, and reproducible DR3 configuration.
- **Extract value-added data products:** homogeneous stellar-continuum and emission-line maps (with uncertainties) plus QC diagnostics for all 26 galaxies.
- **Strong-line O/H baseline + T_e validation:** implement the baseline strong-line metallicity scale for the full sample and measure T_e -based O/H where feasible to validate/refine Z_{gas} .
- **Ionized-gas physics layers:** map DIG contamination and derive ionization-condition proxies and n_e to characterise ISM conditions and test how they shape metallicity and the $Z_{\text{gas}}-\Sigma_{\text{SFR}}$ relation.
- **Environment analysis:** quantify environmental modulation of the $\Sigma_* - \Sigma_{\text{SFR}} - Z_{\text{gas}}$ (and integrated) plane and the mass-dependent inversion signal; identify which environment indicators matter (if any) and where within discs.
- **Write paper:** first-author Project 2 paper presenting the 26-galaxy results, including (i) T_e validation and (ii) quantified dependencies on environment and ISM conditions.

Deliverable: First-author paper extending Project 1 to 26 galaxies, with T_e -based validation of O/H and a quantitative test of environment and ionized-gas-condition modulation.

1.3 Project 3 (future): Add cold gas (MAUVE–ALMA) — testing the gas-supply interpretation

Motivation: The Project 1 regulator framework implies that gas supply and dilution can dominate the sign of the local $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ relation. Project 3 will directly test this by adding molecular gas measurements and asking whether the $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ coupling is better explained (or disappears) once conditioning on the cold-gas reservoir Σ_{H_2} (and related cycling proxies).

Aims:

- **Close the baryon-cycle loop with fuel measurements:** connect Z_{gas} and Σ_{SFR} to Σ_{H_2} (and, where possible, gas-supply-rate proxies).
- **Refine the regulator model:** use molecular gas constraints to calibrate/validate the interpretation of enrichment- vs dilution-dominated regimes.

Core tasks:

- **Homogenise MUSE and ALMA data:** astrometric alignment, resolution matching, common masks/coverage, and consistent uncertainty propagation.
- **Extract H_2 information from ALMA:** derive Σ_{H_2} (and related quantities such as depletion time and gas supply rate) with well-documented assumptions.
- **Gas+metallicity science analysis:** test whether the O/H–SFR coupling changes when controlling for Σ_{H_2} ; evaluate whether molecular gas better traces the “fuel+regulator” variable predicted by the model; explore whether environment effects act indirectly through gas supply.

Deliverable: Paper linking resolved metallicity variability to the molecular gas reservoir and testing the gas-supply/regulator interpretation.

1.4 Indicative timeline (Mar 2026 onward)

- **Mar–Jun 2026 (Project 2):** refine *nGIST* DR3 configuration for value-added products; run all 26 galaxies; deliver homogeneous maps + QC package.
- **Jun–Oct 2026 (Project 2):** refine O/H methodology (DIG/ionization/ n_e ; T_e where feasible); validate that metallicity estimates are robust against environment-correlated systematics.
- **Oct–Dec 2026 (Project 2):** environmental analysis of the $\Sigma_* - \Sigma_{\text{SFR}} - Z_{\text{gas}}$ (and integrated) plane; draft and submit the Project 2 paper.
- **2027 (Project 3):** homogenise MUSE–ALMA products; extract H_2 maps; perform gas+metallicity analysis and regulator-model tests; draft the Project 3 paper.

Timeline

